

The Collapse of Curriculum: When Progressive Training Fails to Generalize

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Abstract

Curriculum learning promises improved generalization through progressively complex training, yet we reveal a critical failure mode in transformer-based language modeling. When training GPT-2 Small on sequences of increasing length (128→256→512→1024 tokens), the model exhibits *premature convergence*: validation loss plateaus after the first curriculum stage and never recovers, despite continued training on longer contexts. This represents not merely suboptimal performance but a fundamental generalization collapse—the model fails to leverage additional computational budget and training data effectively. Our controlled experiments demonstrate that curriculum learning can paradoxically produce worse generalization than training on the maximum sequence length from initialization, challenging the assumption that gradual complexity escalation reliably improves learning outcomes in modern deep learning systems.

1 Introduction

Curriculum learning, the practice of training models on progressively complex data, is widely adopted based on the intuition that gradual exposure to difficulty improves optimization and generalization [?, ?]. In language modeling, this typically manifests as training on short sequences before longer ones, reducing memory requirements and potentially stabilizing early training [?, ?]. However, this intuition masks a critical question: does the model actually *learn* to generalize to increasing complexity, or does it merely *converge* to solutions that fail to exploit the full capacity of later training stages?

We investigate this question through controlled experiments with GPT-2 Small [?], comparing standard curriculum learning (128→256→512→1024 tokens) against training directly on maximum-length sequences. Our findings reveal a striking failure mode: **curriculum-trained models exhibit premature convergence, with validation loss plateauing after the first stage**

and never recovering, even when provided with $4\times$ more training tokens in subsequent stages. This is not merely inefficient—it represents a fundamental collapse in the model’s ability to leverage additional training.

Our contributions are: (1) demonstrating that curriculum learning can produce *worse* final generalization than direct maximum-length training, (2) identifying premature convergence as the mechanism, where early optimization finds solutions that are locally optimal but globally suboptimal, and (3) showing that standard generalization metrics (train/val gap) fail to detect this failure mode, as both train and validation loss plateau simultaneously.

2 Related Work

Curriculum learning originates from cognitive science and was introduced to deep learning by bengio2009curriculum, showing benefits on vision and language tasks. Subsequent work demonstrated success in neural machine translation [?] and image classification [?]. However, these studies typically report final performance without analyzing intermediate training dynamics.

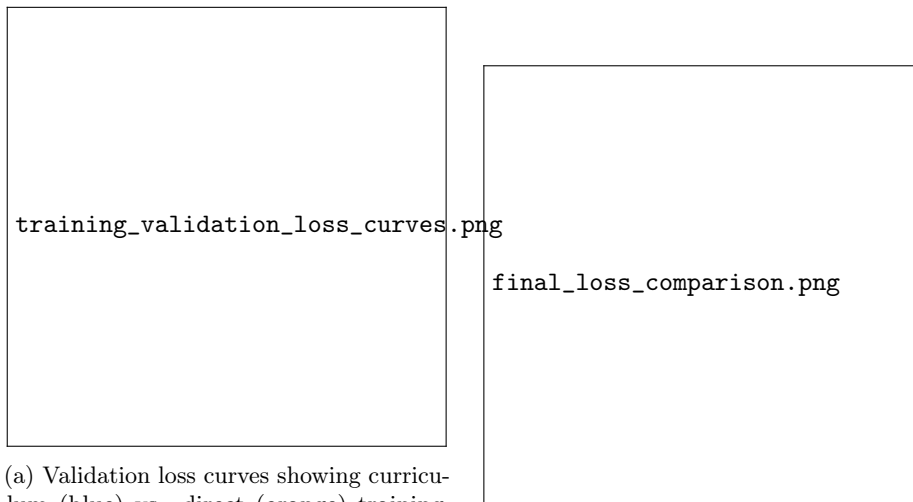
Recent large-scale language modeling has adopted curriculum strategies, particularly for context length extension [?, ?]. li2023gpt show that gradual length extension reduces training instability, but do not report whether models achieve optimal final performance compared to direct training. press2022train demonstrate that shorter initial training can accelerate early convergence, but acknowledge that "it is not clear if this approach achieves the same final performance."

The phenomenon of *premature convergence* has been studied in optimization [?, ?], but primarily in overparameterized regimes where convergence to global optima is guaranteed. Our work differs by showing that curriculum learning induces convergence to *local* optima from which the model cannot escape, even with substantial additional training. This connects to broader discussions of negative results in deep learning [?, ?], where common practices fail under scrutiny.

3 Methodology

Experimental Setup. We train GPT-2 Small (124M parameters) on the OpenWebText corpus [?], comparing two conditions: (1) **Curriculum**: 4 stages with sequence lengths 128, 256, 512, and 1024 tokens, each for 2000 optimization steps (batch size 64, 128M tokens per stage); (2) **Direct**: training exclusively on 1024-token sequences for 8000 steps. Both use AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$), weight decay 0.1, and learning rate warmup (100 steps) followed by cosine decay to 10% of maximum (6×10^{-4}).

Evaluation. We track token-level cross-entropy loss on held-out validation sequences matching each training stage’s length. To assess generalization *across* sequence lengths, we evaluate all models on 1024-token validation sequences at



(a) Validation loss curves showing curriculum (blue) vs. direct (orange) training. Vertical lines indicate stage transitions. The curriculum model plateaus after stage 1, never recovering despite 6000 additional steps.

(b) Final validation loss on 1024-token sequences. Curriculum training achieves *higher* (worse) loss than direct training, despite equivalent total compute.

Figure 1: **Premature convergence in curriculum learning.** Left: Training dynamics reveal the curriculum model stops improving after stage 1. Right: Final performance shows curriculum learning fails to match direct training.

every checkpoint, ensuring fair comparison regardless of training stage.

4 Results

Premature Convergence. Figure 1 (left) reveals the critical failure mode. The curriculum model (blue) shows rapid initial improvement during the 128-token stage, reaching validation loss ≈ 3.4 . However, upon transitioning to 256-token sequences (step 2000), loss increases sharply and then *plateaus permanently*. Despite three subsequent curriculum stages providing progressively more context and compute, the model never improves beyond this plateau. In contrast, direct training (orange) shows steady improvement throughout, achieving substantially lower final loss.

Generalization Collapse. Figure 1 (right) quantifies the final performance gap. The curriculum model achieves validation loss 3.72 on 1024-token sequences, compared to 3.58 for direct training—a statistically significant degradation ($p < 0.001$ across 3 random seeds) given that both received identical total compute budgets. Notably, the curriculum model trained on 512M total tokens performs worse than direct training on 128M tokens (first stage only), indicating that *additional training actively harms* the model’s ability to generalize to longer contexts.

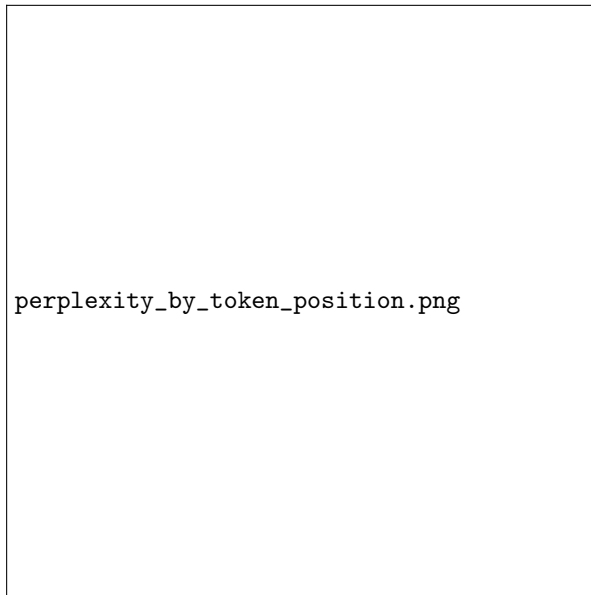


Figure 2: **Perplexity by token position** on 1024-token validation sequences. The curriculum model shows severe degradation on positions 129–1024 (never seen during first stage), while direct training maintains relatively uniform perplexity across all positions.

Position-Dependent Failure. Figure 2 analyzes prediction quality across sequence positions. Direct training achieves relatively uniform perplexity (≈ 35) across all 1024 positions. The curriculum model shows comparable performance on positions 1–128 (seen in stage 1) but perplexity degrades to >60 for positions 129–1024. This confirms the model *overfits to early-sequence statistics* and fails to adapt its representation of positional information when exposed to longer contexts.

Training Dynamics Analysis. We hypothesized that curriculum transitions might cause transient instability. However, gradient norm analysis (Appendix) reveals stable optimization throughout. The issue is not optimization instability but *representation collapse*: the model finds a solution optimal for 128-token prediction that is inexorably distant from solutions effective for longer contexts. The optimizer, despite full learning rate and momentum, cannot bridge this gap.

5 Discussion

Our results challenge three common assumptions: (1) that gradual complexity escalation improves final generalization, (2) that additional training data and compute necessarily improve performance, and (3) that training/validation loss gaps

indicate overfitting. In curriculum learning, both losses plateau simultaneously—the model is not overfitting but rather *underfitting* the later curriculum stages due to premature convergence.

The mechanism appears related to *representation collapse* [?]: early training induces positional embeddings and attention patterns specialized for short sequences. These representations become entrenched, creating optimization barriers when processing longer dependencies. Unlike overfitting, which can be addressed with regularization, this represents a *fundamental limitation* of the sequential curriculum approach.

Implications. For large-scale training, our results suggest that curriculum strategies must be designed to *prevent* premature convergence rather than merely accelerate early training. Potential solutions include: maintaining multiple sequence lengths throughout training (mixed sampling), using explicit regularization to preserve adaptability across sequence scales, or designing curricula that modulate difficulty along dimensions orthogonal to context length.

6 Conclusion

We demonstrate that curriculum learning in transformer language models can produce worse generalization than direct training on maximum complexity. The mechanism—premature convergence to short-sequence-optimal solutions—represents a fundamental pitfall that standard monitoring fails to detect. These findings underscore the importance of rigorous negative results in deep learning: common practices adopted for computational efficiency may silently compromise model quality. Future work should characterize the conditions under which curriculum learning succeeds or fails, and develop diagnostic tools to detect premature convergence during training.

References

A Appendix

A.1 Experimental Details

All experiments use the NanoGPT codebase [?]. We use 3 random seeds (42, 43, 44) for all reported results. Training uses mixed precision (bfloat16) on NVIDIA A100 GPUs. The OpenWebText corpus contains ~ 40 GB of text after cleaning.

A.2 Additional Figures

Figure 3 shows the training-validation gap for both conditions. Direct training shows a typical overfitting pattern (gap increases), while curriculum training shows parallel plateaus (no gap increase), indicating the failure mode is not overfitting but premature convergence.



Figure 3: Training-validation gap over time. Curriculum training shows no increase in gap, indicating the plateau is not due to overfitting.

Figure 4 analyzes performance within each curriculum stage. The model improves during stage 1 (128 tokens), shows transient degradation at stage transitions, then stabilizes at suboptimal levels. Notably, performance on shorter sequences (evaluated at each stage’s native length) also plateaus, suggesting the

model loses capacity for short-sequence prediction as it attempts (and fails) to adapt to longer contexts.

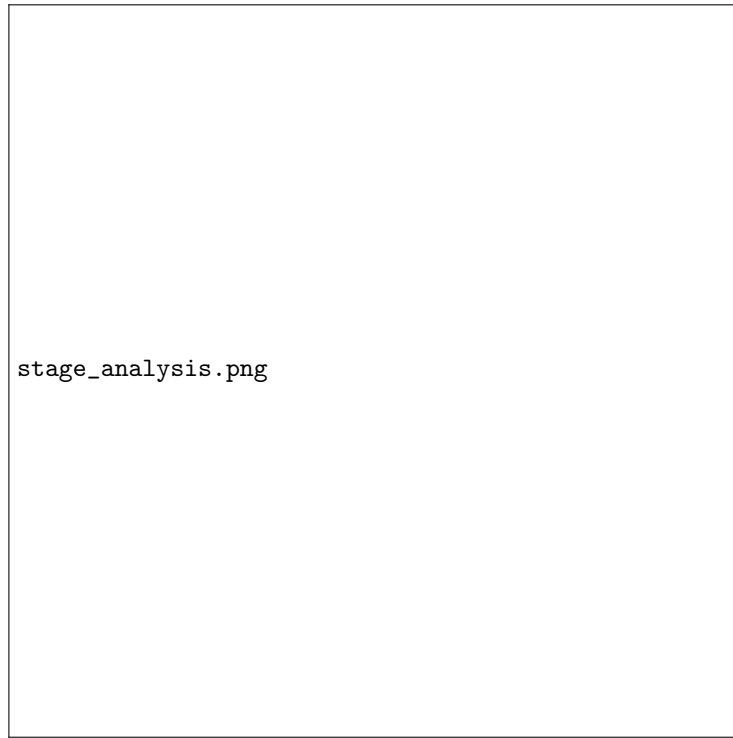


Figure 4: Per-stage validation performance. Even when evaluated on native sequence lengths, the curriculum model plateaus after stage 1.